



Cybersecurity Piscine

Reverse me i'm famous!

Summary: The reverse passion.

Version: 1.00

Chapter I

Introduction

This project aims to discovery the art of Reversing Engineering.

You will learn to understand how several programs work.

For the more adventurous, you will be able to patch these files so that they can validate all the entries you want!

Chapter II

Mandatory part



This project involves using a machine with linux. You can use a VM or whatever you would like to do this project.

- You have 3 programs available on your intra in this project page. There are 3 different difficulties, your goal is just to understand each binary and then find a password to validate each of them.
- You must from this moment understand how theses program works using what you would need (I suggest you to understand how gdb works here).
- You have to write a program in C that will simply be a copy of the basic program.
- Your submission folder must contain at least what is listed below:
 - A folder named by the difficulty of the program.
 - A file that must have the password to pass the crackme in this folder.
 - A file with source.c name containing a representation of C program in this folder.

Your rendering will be of the form:

```
$> ls -al
[...]
```

```
drwxr-xr-x 2 wil wil 4096 Dec 3 XX:42 level1
[...]
```

```
$> ls -alR level1
easy:
total 16
drwxr-xr-x 3 wil wil 4096 Dec 3 XX:42 .
drwxr-xr-x 6 wil wil 4096 Dec 3 XX:42 ..
-rw-r--r-- 1 wil wil XXXX Dec 3 XX:42 password
-rw-r--r-- 1 wil wil XXXX Dec 3 XX:42 source.c
```

```
$> cat level1/password | cat -e  
XXXXXXXXXXXXXXXXXXXXXXXXXXXX$  
$>
```



To validate the mandatory part at the minimal, you only need to reverse the first two programs.

A program is considered valid if a C source representing the algorithmic part of the binary and a valid password to solve this crackme is present in your repository.



It is possible to find several passwords for each binary that is intentional.

Chapter IV

Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defense. Don't hesitate to double check the names of your folders and files to ensure they are correct.